TEAM PROMETHEUS — MISSION DOSSIER & PITCH COMPENDIUM

NASA International Space Apps Challenge 2026 | The Hunt for Planet X | SPHEREx Architecture

Project: Team Prometheus — Planet X Discovery Engine	**Live URL:** www.teamprometheus.us
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1. One-Liner Pitch to Prospective Users

“We turn NASA's SPHEREx infrared sky into a playground where anyone can spot undiscovered worlds beyond Neptune — no telescope or PhD required.”

2. 30-Second Elevator Pitch (Spoken)

“Optical telescopes cannot find Planet X because reflected sunlight drops off by the inverse fourth-power law of distance. Our solution? Don't look for light — detect its heat.

*Team Prometheus built the **SPHEREx Dual-Engine Architecture**. We use 102 continuous infrared bands to power **Citizen Explorer**, where moving planets pop out through automated differential blinking, and **Scientist Mode**, which filters 1.8 billion Gaia background stars and fits Spectral Energy Distribution curves.*

*Zero server costs. Fully live right now at **teamprometheus.us**!”*

3. 60-Second Competition Pitch (Spoken — Plain Words, No Formulas)

“In nineteen thirty, Clyde Tombaugh spent seven thousand hours manually examining photographic glass plates to discover Pluto. Today, finding Planet X with optical telescopes is nearly impossible because reflected sunlight drops off by the inverse fourth-power law of light.

Our solution? Don't search for reflected sunlight — detect the planet's internal thermal heat. NASA's upcoming SPHEREx observatory scans the sky across one hundred and two continuous infrared channels.

*I'm Diwakar, and alongside Swapnil Kumar, Team Prometheus built the **SPHEREx Dual-Engine Architecture**.*

*For the public, **Citizen Explorer** turns infrared data into visual false-color composites with automated differential blinking so anyone can spot moving objects. For researchers, **Scientist Mode** filters out one point eight billion catalog stars and fits thermal blackbody curves.*

*Zero server cost, zero installs. It is fully live at **teamprometheus.us**. Help us hunt for Planet X!”*

4. Why I Chose This Challenge (Submission Answer)

Official Written Submission Text:

We chose this challenge because it addresses one of the greatest unsolved mysteries in modern astronomy: **the outer frontier of our own solar system**.

Humanity has discovered over five thousand exoplanets orbiting distant stars hundreds of light-years away, yet the farthest reaches of our own cosmic backyard beyond Neptune remain uncharted. Gravitational anomalies in the orbits of extreme trans-Neptunian objects strongly suggest a massive fifth giant planet exists, but optical telescopes are blinded by distance.

With NASA preparing to launch the **SPHEREx observatory**, astronomy is on the verge of receiving an unprecedented, all-sky hyperspectral infrared survey across one hundred and two continuous bands. We saw a rare opportunity to build the software pipeline before the data arrives, solving the historical bottleneck that forced Clyde Tombaugh to spend seven thousand hours manually inspecting glass plates in nineteen thirty.

Most importantly, we chose this challenge to **democratize cosmic discovery**. Big-data astrophysics is traditionally locked behind supercomputing clusters and specialized software. We wanted to prove that modern web engineering can bring calibrated NASA science directly into any browser with zero server costs — allowing students, citizen explorers, and professional researchers to collaborate on uncovering the next planet in our solar system.

Quick 30-Second In-Person Spoken Version:

> *“We've discovered thousands of planets around distant stars, but our own solar system's outer edge is still an unmapped frontier. Gravitational evidence points to a hidden planet, but optical telescopes can't see it.*
> *With NASA's upcoming SPHEREx mission capturing the entire sky across one hundred and two infrared channels, we chose this challenge to build the discovery engine ready on day one.*
> *Instead of locking this data behind supercomputers, we built a zero-cost tool where any curious citizen or astrophysicist can spot a moving planet directly from their browser.”*

5. How We Will Reach Users & Scaling Strategy

1. NASA's Built-In Distribution Network

NASA Space Apps Challenge is the world's largest annual hackathon, reaching over 150 countries and 50,000+ participants. Winning or placing gives our platform direct visibility across NASA's official channels, social media, and partner networks as the primary launchpad.

2. Zero Barrier to Entry

Our platform runs entirely in the browser at teamprometheus.us. No downloads, no installations, no accounts, no payments. Anyone with a phone, tablet, or laptop and an internet connection can start exploring immediately.

3. Dual-Audience Strategy

Citizen Explorer Mode targets students, science enthusiasts, and curious people with zero technical background using gamified visual blinking.
Scientist Mode targets university researchers and amateur astronomers needing calibrated data and peer-review-ready dossiers.

4. Education & Classroom Adoption

Partnering with high school physics and university astrophysics programs. Teachers can assign 'hunt for a candidate' as an interactive classroom lab where students interact with real infrared astronomical data.

5. Astronomy Communities & Social Media

Targeting Reddit (r/astronomy, r/space), Discord astronomy servers, and citizen science hubs like Zooniverse. Short viral clips showing differential blinking in action demonstrate discovery in real time.

6. Open-Source & Community Contributions

The entire codebase is open on GitHub. Developers can add features, astronomers can contribute pipelines, and educators can build customized modules.

Spoken Summary for Judges:

> *“First, the NASA Space Apps platform gives us day-one global visibility. Second, our zero-barrier browser app means anyone on any device can start hunting in five seconds with no install. Third, we target both sides: science teachers looking for interactive classroom labs, and online astronomy communities hungry for real discovery. We don't just build a tool — we build a community of planet hunters.”*

6. Project Solution (Under 500 Words)

The Problem: Hunting for an elusive 5–10 Earth-mass planet in the outer solar system (400–800 AU) is fundamentally limited by the optical reflection barrier. Reflected sunlight drops off with the fourth power of distance, making Planet X over 100 billion times fainter than at 1 AU. In 1930, Clyde Tombaugh spent 7,000 hours manually blinking glass plates. Repeating this across modern terabyte-scale surveys is impossible without automated pipelines.

The Solution: Team Prometheus shifts the detection paradigm from optical reflection to **infrared thermal emission**. While reflected optical flux is virtually undetectable, a core-warmed 40 K world peaks in the mid-infrared. NASA's SPHEREx mission maps the entire celestial sphere across 102 continuous spectral bands (0.75–5.0 μm). We built the **SPHEREx Dual-Engine Discovery Architecture**, fully deployed and running client-side at **teamprometheus.us**.

Architecture Overview:

• **Citizen Explorer Engine:** Designed for public discovery. Synthesizes 102 hyperspectral channels into false-color RGB composites. Features an automated Differential Blink Comparator mimicking Tombaugh's method at 2 Hz, isolating orbital proper motion (1.5–3.0 arcseconds/year) against stationary stellar backgrounds.

• **Scientist Mode Engine:** For professional astrophysics. Cross-matches candidate detections against 1.8 billion stars in the Gaia DR3 catalog, rejecting stationary interlopers down to magnitude 21. Computes multi-epoch astrometry, estimates orbital semimajor axis, and fits blackbody SED curves to identify characteristic methane (1.66 μm, 2.3 μm) and water-ice (3.05 μm) atmospheric absorption.

• **Autonomous Verification Agent:** A localized AI assistant analyzing candidate signal-to-noise ratio, background star density, and spectral plausibility, generating peer-review-ready dossiers.

Impact: Completely serverless, zero operating costs, running in any modern browser. It democratizes the search for the solar system's next planet.

7. Comprehensive Solution Dossier (Under 3,000 Characters)

The Challenge: The Optical Reflection Barrier
Hunting for an elusive massive planet in the outer solar system (between four hundred and eight hundred astronomical units from the Sun) is fundamentally limited by the inverse fourth-power law of optical illumination. Reflected sunlight from an object at five hundred astronomical units is over one hundred billion times fainter than near Earth. In nineteen thirty, Clyde Tombaugh spent seven thousand hours manually inspecting photographic glass plates to discover Pluto. Today, brute-force visual scanning across terabytes of survey data is impossible without intelligent software architectures.

The Paradigm Shift: From Optical Reflection to Thermal Emission
Team Prometheus bypasses the optical barrier by tracking the planet's internal thermal radiation. A planet with a core-warmed effective temperature of approximately forty Kelvin emits significant thermal radiation peaking in the mid-infrared spectrum. NASA's upcoming SPHEREx all-sky survey maps the cosmos across one hundred and two continuous infrared spectral bands from point seven five to five microns. We engineered the **SPHEREx Dual-Engine Discovery Architecture**, a web-native platform deployed live at **teamprometheus.us**.

Dual-Engine Architecture:
1. **Citizen Explorer Mode:** Transforms complex hyperspectral data cubes into intuitive visual discoveries. Generates false-color composites and runs an automated Differential Blink Comparator at two hertz, making slow trans-Neptunian orbital motion immediately apparent to any user.
2. **Scientist Mode:** Provides a complete research workspace. Automatically queries the Gaia astronomical catalog to filter out stationary background stars, calculates candidate proper motion, estimates orbital distance, and performs blackbody Spectral Energy Distribution curve fitting to identify methane (one point six six and two point three microns) and water ice (three point zero five microns) absorption bands.
3. **Autonomous Astrophysics Agent:** Evaluates signal-to-noise ratio and false-positive likelihood to produce verified, publication-ready candidate dossiers.

Real-World Impact: Zero server overhead, zero software installation, and instantaneous browser accessibility. Team Prometheus turns NASA's premier infrared mission into a collaborative discovery engine for both the public and researchers.

8. Scientific Feasibility & Honest Assessment

Can this system really help discover new planets?
• **The Science is Sound:** Planet Nine is a legitimate astronomical hypothesis backed by orbital clustering of distant TNOs (Batygin & Brown). Infrared detection is the correct physical approach because cold outer planets glow in thermal IR while remaining invisible in optical light.
• **Citizen Science has Precedent:** Zooniverse's Planet Hunters and Backyard Worlds: Planet 9 successfully found exoplanets and brown dwarfs using human pattern recognition paired with WISE and Kepler data.
• **Prototype to Discovery Pipeline:** The hackathon platform proves that 102-band infrared false-color rendering and differential blinking work flawlessly in a web browser. With multi-epoch SPHEREx survey releases, connecting real epoch-to-epoch astrometric motion vectors turns this architecture into a genuine discovery engine.